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ADVANCEMENT IN THE FIELD OF NANO-TECHNOLOGY

INTRODUCTION

Nanotechnology is molecular manufacturing or more simply, building things the size of one atom or one molecule with programmed precision. It involves working with matter at the scale of one-billion of a meter (**Nano,** symbol-n, is a unit prefix meaning "one billionth". Used primarily with the metric system, this prefix denotes a factor of 10^{-9} or 0.000000001). The idea of nanotechnology is, therefore, to master the characteristics of matter intelligently to develop a highly efficient system. It is an interesting confluence of biology, chemistry and electronics. The trick is to manipulate atoms individually and place them where needed, to produce the desired structure. Therefore, it is often termed as the 'bottom-up' manufacturing. It is used for areas of science and engineering where phenomena which take place at nanometer level are utilized in the design, characterization, production and application of materials, structures, devices and systems.

Although there are many examples of structures in nature, which exist at nanometer dimensions like essential molecules with the human body and components of food, and although many technologies have incidentally involved nanoscale structure for many years, it has only been in the last quarter of the twentieth century that it has been possible to actively and intentionally manipulate molecules and structures at nano level. It is this control at the nanometer scale level that distinguishes nanotechnology from other areas of technology.

The National Nanotechnology initiative **(**The National Nanotechnology initiative is a United States federal government program for the science, engineering, and technology research and development for nanoscale projects) defines nanotechnology as the manipulation of matter with atleast one dimension sized from 10 to 200 manometres.

As per UK Royal Society and Royal Academy of Engineering report, 2004 - The range of the nanoscale is from the atomic level, at around 0.2 nm up to around 100

nm. It is within this range that materials can have substantially different properties compared to the same substances at larger sizes both because of the substantially increased ratio of surface area to mass, and also because quantum effects begin to play a role at these dimensions, leading to significant changes in several types of physical property.

APPLICATIONS OF NANOTECHNOLOGY:

Nanotechnology has numerous applications:

1. **NANOTECHNOLOGY IN MEDICINE:**
    1. One application of nanotechnology in medicine currently being developed involves employing nanoparticles to deliver drugs, heat, light or other substances to specific types of cells (such as cancer cells). Particles are engineered so that they are attracted to diseased cells, which allows direct treatment of those cells.
    2. This technique reduces damage to healthy cells in the body and allows for earlier detection of disease. For example, nanoparticles that deliver chemotherapy drugs directly to cancer cells are under development. Tests are in progress for targeted delivery of chemotherapy drugs and their final approval for their use with cancer patients is pending.
    3. Nanoparticles that release drugs when subjected to sheer force, such as occur when passing through a section of artery that is mostly blocked by a clot. This method is effective in delivering drugs used to dissolve clots.
    4. Gelatin nanoparticles can be used to deliver drugs to damaged brain tissue more efficiently that standard methods. This method will result in more effective drug delivery for brain injuries.
    5. A sponge-like matrix that contains insulin as well as Nano capsules containing Hydrogen ions. When the glucose level rises the Nano capsules release hydrogen ions, which bind to the fibres making up the matrix. The hydrogen ions make the fibres positively charged, repelling each other and creating openings in the matrix through which insulin is released. This procedure is more effective as it releases insulin only when the glucose levels are high.

6. A nanoparticle can also be used that can be taken orally and pass through the lining of the intestines into the bloodstream.
7. Nano particles can also be use that deliver an enzyme that prevent the reproduction of viruses molecules in the patients' bloodstream.
8. 'Nano sponges' that absorb toxins and remove them from the bloodstream. The Nano sponges are polymer nanoparticles coated with a red blood cell membrane. The red blood cell membrane allows the Nano sponges to travel freely in the bloodstream and Nano sponges attract the toxins.
9. Bismuth nanoparticles are used to concentrate radiation in radiation therapy to treat cancer tumours. Initial results indicate that the bismuth nanoparticles would increase the radiation dose to the tumor by 90 percent.
10. Targeted heat therapy is being developed to destroy breast cancer tumors. In this method antibodies that are strongly attracted to proteins produced in one type of breast cancer cell are attached to nanotubes, causing the nanotubes to accumulate at the tumor. Infrared light from a laser is absorbed by the nanotubes and produces heat that incinerates the tumor.
11. Microbivores: Nano-sized artificial Mechanical Phagocytes that digest the infectious bacteria in the patient's body are also an important application.

2. NANOTECHNOLOGY IN ELECTRONICS: NANOELECTRONICS

Nanoelectronics holds some answers for how we might increase the capabilities of electronic devices while we reduce their weight and power consumption. Some of the applications are given below.

1. Integrating silicon nanophotonics components into CMOS (Complementary metal oxide semiconductor) integrated circuits. This optical technique is intended to provide higher speed data transmission between integrated circuits than is possible with electrical signals.
2. A low power method to use nanomagnets as switches, like transistors, in electrical circuits is also into consideration for development. This method might lead to electrical circuits with much lower power consumption than transistor-based circuits.

3. Another idea is to print prototype circuit boards using standard inkjet printers to form the conductively lines needed I circuit boards.
4. Building transistors from carbon nanotubes to enable minimum transistor dimensions of a few nanometres and developing techniques to manufacture integrated circuits built with nanotube transistors.
5. Using electrodes made from nanowires that would enable flat panel displays to be flexible as well as thinner than current flat panel displays.
6. Using semiconductor nanowires to build transistors and integrated circuits.
7. Using carbon nanotubes to direct electrons to illuminate pixels, resulting in a lightweight, millimetre thick 'nenoemmissive' display panel.
8. Using nanosized magnetic rings to make Magnetoressistive Random Access Memory (MRAM) which research has indicated may allow memory density of 400 GB per square inch.

3. NANOTECHNOLOGY IN FOOD

Contamination Sensor: Flash light to reveal the presence of E.Coli bacteria.

Antimicrobial Packaging: Edible food films made with cinnamon or oregano oil, or nano particles of zinc, calcium other materials that kill bacteria.

Improved Food Storage: Nano-enhanced barrier keeps oxygen-sensitive foods fresher.

Enhanced Nutrient Delivery: Nano-encapsulating improves solubility of vitamins, antioxidants, healthy omega oils and other 'nutraceuticals'.

Green Packaging: Nano-fibers made from lobster shells or organic corn are both antimicrobial and biodegradable.

Pesticide Reduction: A cloth saturated with nano fibres slowly releases pesticides, eliminating the need for additional spraying and reducing chemical leakage into the water supply.

Tracking, Tracing; Brand Protection: Nanobarcodes can be created to tag individual products and trace outbreaks.

Texture: Food spread ability and stability improve with nano-sized crystals and lipids for better low-fat foods.

Flavour: Trick the tongue with bitter blockers or sweet and salty enhancers.

Bacteria Identification and Elimination: Nano carbohydrate particles bind with bacteria so they can be detected and eliminated.

1. Storage bins are being produced with silver nanoparticles embedded in the plastic. The silver nanoparticles kill bacteria from any material that was previously stored in the bins, minimizing health risks from harmful bacteria.
2. Using silicate nanoparticles to provide a barrier to gasses (for example oxygen), or moisture in a plastic film used for packaging. This could reduce the possibility of food spoiling or drying out.
3. Pesticides encapsulated in nanoparticles; that only release pesticide within an insect's stomach, minimizing the contamination of plants themselves.
4. Another development being pursued is a network of nano sensors and dispensers used throughout a farm field. The sensors recognize when a plant needs nutrients or water before there is any sign that the plant is deficient. The dispensers then release fertilizer, nutrients, or water as needed, optimizing the growth of each plant in the field one by one.

4. AGRICULTURE

NANOTECHNOLOGY FOR CROP BIOTECHNOLOGY

Nanoparticles can serve as magic bullets, containing herbicides, chemicals, or genes, which target particular plant parts to release their content. Nano capsules can enable effective penetration of herbicides through cuticles and tissues, allowing the slow and constant release of the active substances.

NANOTECH DELIVERY SYSTEMS FOR PESTS, NUTRIENTS, AND PLANT HORMONES

1. Nano-sensors and nano-based smart delivery systems could help in the efficient use of agricultural, natural resources like water, nutrients and chemicals through precision farming. Through the use of nanomaterials and global positioning systems with satellite imaging of fields, farm managers could remotely detect crop pests or evidence of stress such as drought. Once pest or-drought is detected there would be an automatic adjustment of pesticides applications or irrigation levels.

2. Nano-sensors dispersed in the field can also detect the presence of plant viruses and the level of soil nutrients. Nano-encapsulated slow release fertilisers have also become a trend to save fertilizers consumption and to minimise environmental pollution.
3. Nano-barcodes and nano-processing could also be used to monitor the quality of agricultural produce. Scientists at Cornell University used the concept of grocery barcodes for cheap, efficient rapid and easy decoding and detection of diseases.
4. They produced microscopic probes or nano-barcodes that could tag multiple pathogens in a farm which can easily be detected using any fluorescent-based equipment. This ongoing project generally aims to develop a portable onsite detector which can be used by non-trained individuals.
5. Through nanotechnology, scientists are able to study plant's regulation of hormones such auxin which is responsible for root growth and seeding establishment.

5. NANOTECHNOLOGY IN FUEL CELLS

1. It can reduce the amount of platinum needed in the fuel cells.
2. The storage of hydrogen in graphene (single atom thick carbon sheets). The researchers found that they could store 14% by weight of hydrogen in graphene.

6. NANOTECHNOLOGY IN SPACE

1. Employing materials made from carbon nanotubes to reduce the weight of spaceships.
2. Including layers of bio-nano robots in spacesuits. The outer layer of bio-nano robots would respond to damages to the spacesuit, for example, to seal up punctures. An inner layer of bio-nano robots could respond if the astronaut were in trouble, for example by providing drugs in a medical emergency.
3. Deploying a network of nanosensors to search large areas of planets such as Mars for traces of water or other chemicals.
4. Using carbon nanotubes to build lightweight solar sails that use the pressure of light from the sun reflecting on the mirror-like solar cell to propel a spacecraft.

This solves the problem of having to lift enough fuel into orbit to power spacecraft during interplanetary missions.

5. Working with nanosensors to monitor the levels of trace chemicals in spacecraft to monitor the performance of life support systems.

7. NANOTECHNOLOGY IN MAKING EFFICIENT FUELS

Nanotechnology, in life form of genetic engineering, can also improve the performance of enzymes used in the conversion of cellulose into ethanol.

8. NANOTECHNOLOGY IN REDUCING AIR POLLUTION

There are two major ways in which nanotechnology is being used to reduce air pollution.

1. Nanotechnology can improve the performance and cost of catalysis used to transform vapour escaping from cars or industrial plants into harmless gasses. That's because catalysis made from nanoparticles have a greater surface area to interact with the reacting, which makes the catalyst more effective.
2. Nanostructured membranes are being developed to separate carbon dioxide from industrial plant exhaust streams.

9. NANOTECHNOLOGY IN REDUCING WATER POLLUTION

Nanotechnology is being used to develop solutions to different problems in water quality.

1. Nanoparticles can be used to convert the contaminating chemicals in the industrial water through a chemical reaction to make it harmless. Studies have shown that this method can be used successfully to reach contaminates dispersed in underground ponds and at a much lower cost than methods which require pumping the water out of ground for treatment.
2. A do-ionisation method using electrodes composed of nano-sized fibres can-be-used for reducing the cost and energy requirements of turning salt water into drinking water.
3. A filter only a few nanometres, in diameter is currently being developed that should eb capable of removing viruses from water.

4. Using nano scavengers, in which a layer of reactive nanoparticles coats a synthetic core which is designed to be easily magnetized. The nanoparticles, for example, silver nanoparticles attach to and kill the bacteria. Then when a magnetic field is applied the nano scavengers are removed from the water.
5. Using pellets containing nano structured palladium and gold as a catalyst to breakdown chlorinated compounds contaminating groundwater. Since palladium is very expensive, the researchers formed the pellets of nanoparticles that allow almost every atom of palladium to react with the chlorinated compounds, reducing the cost of the treatment.
6. Carbon nanotubes are being developed to clean up oil spills. Researchers have found that adding boron atoms during the growth of carbon nanotubes causes the nanotubes to grow in a sponge-like material that can absorb many times its weight in oil.

10. NANO TECHNOLOGY TO IMPROVE BATTERIES

1. Reducing the possibility of batteries catching fire by providing less flammable electrode material.
2. Increasing the available power from a battery and decreasing the time required to recharge a battery by coating the surface of an electrode with nanoparticles.
3. Increasing the shelf life of a battery by using nanomaterials to separate liquids in the battery from the solid electrodes when there is no draw on the battery. This separation prevents the low-level discharge that occurs in a conventional battery, which increases the shelf life of the battery dramatically.
4. Use of silicon nanoparticles-based anodes can increase the capacity of Li-ion batteries by up to 10 times.
5. Ultra-capacitor using single atom thick graphene sheets to store electrical change.

MORE USES OF NANO-TECHNOLOGY

CLOTHES THAT CLEAN THEMSELVES

A group of researchers located in Australia at Monash University in Victoria are developing a process that could lead to self-cleaning wool and silk. They have found a

way to coat fibres with titanium dioxide nano-crystals, which break down food and dirt in sunlight.

PHOTOLITHGRAPHY AND NANOLITHOGRAPHY

Photolithography is a process used in micro fabrication which is the removal of thin films selectively (or the bulk of a substrate), by utilizing a light to transfer a geometric pattern from a photo mask to a light-sensitive chemical on the substrate. To develop modern microprocessors, manufacturers use 'photolithography', the highest technology in the nano-scale dimension of printing technology Photolithography has been a perfect way to produce massively integrated circuits and computer chips.

NANOMATERIALS

The nanomaterials field includes subfields which develop or study materials having unique properties arising from their nanoscale dimensions. Nanomaterials are made-possible by manipulating matter at the nanoscale (less than 100 nanometres, or approximately molecular scale).

At the nanoscale, ordinary substances like carbon and clay take on surprising properties – including greater reactivity, unusual electrical properties, and enormous strength per unit of weight – that can enable the creation of new types, of medicine, solar cells, water desalination and filtration, super-stick coatings, and stronger composites. Invisible to the naked eye, nanomaterials have already found their way into products as varied as pharmaceuticals, sunscreens, bacteria-killing socks, and composites bicycle frames. And recent generations of semiconductors have featured in life tens of nanometers, in effect making them nanotechnology. Research has given rise to many materials which may be useful in nanotechnologies, such as carbon nanotubes and other fullerenes, and various nanoparticles and nanorods.

NANOROBOTICS

It is the technology of creating robots or nanomachines at the microscopic scale of nanometers (10-9 meters) or at least very close to this scale. The nanorobots are also named 'nanobots', 'nanorods' of 'nanites'. Scientists have been developing these robots to carry out many important functions in different professional fields such as medicine,

engineering, manufacturing and many others even though non-biological nanorobots are yet to be created.

Optical uses

Advances have been made to produce the first anti-reflective ultrathin polymer sunglasses. In optical surgery, nano-optics increases the precision of pupil repair, and other types of surgery using lasers technology.

THE CHALLENGES, RISKS AND ETHICAL Issues with nanotechnology

The most immediate challenge in nanotechnology is that we need to learn more about materials and their properties of the nanoscale. Universities and corporations across the world are rigorously studying how atoms fits together to form larger structures. We're still learning about how quantum mechanics impact substances at the nanoscale. Because elements at the nanoscale behave differently than they do in their bulk form, there's a concern that some nanoparticles could be toxic. Some doctors worry that the nanoparticles are so small, that they could easily cross the blood-brain barrier, a membrane that protects the brain from harmful chemicals in the bloodstream. If we plan on using nanoparticles to coat everything from our clothing to our highways, we need to be sure that they won't poison us.

There are some argumentations reforming to thinking of the social impact of nanotechnology as a technologic an impact that will change dramatically the fabric of society. People dealing with the ethical aspects of nanotechnology argue that such transformative technology could exacerbate the division of rich and poor known as 'nano-divide'. He critics of nanotechnology point to the potential toxicity of new classes of nano substances that adversely affect the stability of cell membranes or disturb the immune system when inhaled or digested. The titanium oxide nanoparticle used in sunscreen can get into the human food chain, which causes quicker ageing of colon cells. The cadmium selenide nanoparticles, also called quantum dots, can cause cadmium poisoning in humans.

The Centre for Responsible Nanotechnology warns of the broad societal implications of untraceable weapons of mass destruction, networked camera for use by the government, and weapons developments fast enough to destabilize arms races.

Another area of concern is the effect that industrial-scale manufacturing and use of nanomaterials would have on human health and the environment, as suggested by nanotoxicology research. For these reasons, groups such as the Centre for Responsible Nanotechnology advocate that nanotechnology be regulated by governments. Others counter that overregulation would stifle scientific research and the development of beneficial innovations. Some nanoparticle products may have unintended consequences. Researchers have discovered that bacteriostatic silver nanoparticles used in socks to reduce foot odour are being during the wash. These particles are then flushed into the waste water stream and may destroy bacteria which are critical components of natural ecosystems, farms, and waste treatment processes.

In general, smaller particles are more bioactive and toxic. Their ability to interact with other living systems increases because they easily get into the body, and there may be further biochemical reactions like the creation of free radicals that damage cells. There is no doubt that nano-particles have interesting and useful properties strong economic drives and competition in the market place may be taking precedence over methodical scientific prudence when it comes to public health and possible dangers of nano technology.

There are also fears that the environment may be placed in jeopardy that nanotechnology products may increase the pollution level in many areas at a time when the need for environmental conservation is vigorously pursed everywhere. For example, the production of better engines for automotive and windows for cars may prompt car manufacturers and local dealers to wantonly discard these parts that can possibly cause an environmental nightmare with the absence of better program on waste disposal. Because nano particles are very small, they are very difficult to degrade and may easily pass on from one person to another to the process becoming agents of ailments.

If nanotechnology in medicine makes it possible for us to enhance ourselves physically, is that ethical? In theory, medical nanotechnology could make us smarter, stronger and give us other abilities ranging from rapid healing to night vision. Should we pursue such goals? Could we continue to call ourselves human, or would we become transhuman – the next step on man's evolutionary path? Since almost every technology

starts off as very expensive, would this mean we'd create two races of people. A wealthy race of modified humans and a poorer population of unaltered people? We don't have answers to these questions, but several organizations are urging nano scientists to consider these implications now before it becomes too late. Many experts think that concerns like grey goo and trans humans are at best premature, and probably unnecessary. Even so nanotechnology will definitely continue to impact us as we learn more about the enormous potential of the nanoscale'.

NANOTOXICOLOGY

Nanotoxicology is the study of the toxicity of nanomaterials. Because of quantum size effects and large surface area to volume ratio; nanomaterials have unique properties compared with their larger counterparts. Nanotoxicology is branches of bio nanoscience which deals with the study and application of toxicity of nanomaterials. Nanomaterials, even when made of inert element like gold, become highly active at nanometer dimensions. Nanotoxicological studies are intended to determine whether and to what extent these properties may pose a threat to the environment and to human beings. For instance, Diesel nanoparticles have been found to damage the cardiovascular system in a mouse model.

NANOPOLLUTION

Nanopollution is a genetic name for all waste generated by nanodevices or during the nanomaterials manufacturing process. This kind of waste may be very dangerous because of its size. It can float in the air and might easily penetrate animal and plant cells causing unknown effects. Most human made nanoparticles do not appear in nature, so living organisms may not have appropriate means to deal with nanowaste. It is probably one great challenge to nanotechnology: how to deal with its nano pollutants and nanowaste.

NANOTECHNOLOGY RESEARCH IN INDIA

A vision group under the leadership of Prof. CNR Rao was formed and entrusted with the task of formulating a National Nanotechnology Policy. Under this $250 million has been allocated for R &D in Nanotechnology. Two centres of excellence, i.e. Institute of nanoscience and nanotechnology have been proposed to be built in Mumbai and

Bangalore. Sabeer Bhatia, co-founder of Hotmail, has proposed to build a nano-city in Chandigarh.

Nano Science and Technology Initiative (NSTI)

Nanoscience and technology are receiving increasing attention all over the world. The department of science and technology of government of India has assessed the importance of this highly emerging inter-disciplinary field and launched a national programme titled nano-science and technology initiative" (NSTI). The programme focused at overall research and development in nano science and technology with vigour so that India can becomes significant player in the area and contribute to the development of new technologies besides carrying out basic research. Under this initiative, close to 100 research projects on the synthesis and assembly of ceramic, nanoparticles, nanotubes, nanowires, nanopores, solids, nanostructured alloys and DNA chips have been supported. An impressive array of instruments has also been put in place to facilitating characterization of the synthesized materials. On the application side, projects on nanolithography, nanostructured high strength materials, targeted drug delivery systems, DNA chips etc., have been supported. Mine centre of excellence has been established in different regions of the country.

The Department of Information Technology has started Nanotechnology Development Programme to create infrastructure for research in nanoelectronics and nano-meteorology at National level and also to fund small and nano devices, carbon nano lies (CTN) nano systems etc. The Department has initiated a major project for setting up of nano-electronics centres at IISC, Bangalore and IIT, Bombay.

International Center for Technology/Assessment (ICTA)

ICTA is a U.S. non-profit bi- partisan organization, based in Washington, D.C. ICTA arms to provide the public with full assessment's and analyses of technological impacts on society. ICTA explores the economic, ethical, social, environmental and political impacts that can result from the applications of technology or technological systems.

ICTA seeks to assist the public and policy makers in better understanding how technology affects society. ICTA works towards adequate oversight of nanotechnology

through its Nanotechnology Project NanoAction. The implications of nanotechnology run the gamut from the medical, ethical, mental, legal and environmental applications, to fields such as engineering, biology, chemistry, computing, materials science, military applications, and communications.

Questions:

1. Discuss what are the advancements in the field of Nano-Technology in India.
2. Describe how the Nano-Technology applied in Food and Agriculture Sector.
3. Give a short note on
   a) Nanorobotics
   b) Photolithography
   c) Nanotoxicology.